Lecture Notes

Lecture 7

Inner Product Spaces

So far our vector spaces have had no geometry: no length, no angle, no notion of two vectors being perpendicular. The *inner product* supplies all three. You know it on $\mathbb{R}^2$ and $\mathbb{R}^3$; now we define it on any vector space, including a complex one. In quantum mechanics the squared modulus of the inner product of two unit state vectors is a probability, and the main result of the lecture, the Cauchy–Schwarz inequality, shows that it never exceeds 1. Lectures 7–10 work with finite-dimensional spaces over $\mathbb{R}$ or $\mathbb{C}$; function spaces appear only as examples.

We use the *physics convention*: the inner product is conjugate-linear in the first slot and linear in the second. Many mathematics texts put the conjugation in the second slot; their formulas convert by swapping the two arguments.

Learning objectives

By the end of this lecture you will be able to:

- Verify the inner-product axioms in the physics convention.
- Compute norms, check orthogonality, and apply the Pythagorean theorem.
- Prove and apply the Cauchy–Schwarz inequality and the triangle inequality.
- Recover the inner product from the norm by the polarization identity.
- Translate between the notation $\langle \cdot, \cdot \rangle$ and Dirac's notation $\langle \cdot \mid \cdot \rangle$.

1 Inner products

Definition 7.1 (Inner product, physics convention). An *inner product* on a complex vector space V assigns to each ordered pair $u, v \in V$ a scalar $\langle u, v \rangle \in \mathbb{C}$ such that

1. *conjugate symmetry*: $\langle u, v \rangle = \overline{\langle v, u \rangle}$;
2. *linearity in the second slot*: $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle$;
3. *positive definiteness*: $\langle v, v \rangle \geq 0$, with equality iff $v = 0$.

A vector space with an inner product is an *inner product space*.

By conjugate symmetry $\langle v, v \rangle = \overline{\langle v, v \rangle}$, so the number $\langle v, v \rangle$ is real. Hence the positive-definiteness condition makes sense. This is why the first slot carries a conjugation: without it $\langle v, v \rangle$ could be complex, and we could not define the length of a vector. Combining conjugate symmetry with linearity in the second slot gives *conjugate-linearity in the first slot*:

$$\langle au + bw, v \rangle = \overline{\langle v, au + bw \rangle} = \overline{a\langle v, u \rangle + b\langle v, w \rangle} = \overline{a}\,\langle u, v \rangle + \overline{b}\,\langle w, v \rangle.$$

So scalars come out of the first slot conjugated and out of the second slot unchanged. Such a product is "one and a half" times linear, and is called *sesquilinear*. Over the reals the conjugation changes nothing. So a real inner product is a symmetric, positive-definite *bilinear* form, that is, one that is linear in each slot separately. Bilinear forms are studied in their own right in Lecture 11.

Example 7.2 (Examples of inner products).

(1) On $\mathbb{C}^n$, the *standard* inner product is

$$\langle u, v\rangle = \sum_{k=1}^{n} \overline{u_k}\, v_k = u^\dagger v.$$

The coordinates of the *first* vector are conjugated, and $u^\dagger = \overline{u}^{\mathsf{T}}$ is the conjugate transpose (with u written as a column). On $\mathbb{R}^n$ the conjugation changes nothing, and we get the familiar dot product $\sum_k u_k v_k$.

(2) On the space of continuous functions on $[a, b]$, with $a < b$,

$$\langle f, g\rangle = \int_a^b \overline{f(t)}\, g(t)\, dt$$

is an inner product. In quantum mechanics, the states of particles are described in similar function spaces.

(3) On $M_{n\times n}(\mathbb{C})$, the *Frobenius* inner product is

$$\langle A, B\rangle = \operatorname{tr}(A^\dagger B) = \sum_{j,k} \overline{A_{jk}}\, B_{jk}.$$

It is the standard inner product on $\mathbb{C}^{n^2}$ in disguise: write all n^2 entries of the matrix as one long vector. Later we will use it to measure the "size" of operators.

(4) With fixed weights $w_1, \dots, w_n > 0$, the *weighted* product $\langle u, v\rangle = \sum_k w_k \overline{u_k} v_k$ is also an inner product on $\mathbb{C}^n$. The weights must be positive, because $\langle e_k, e_k\rangle = w_k$ for the standard basis vector e_k; Exercise 7.2 shows what goes wrong otherwise. Positive definiteness is what makes $\|v\| = \sqrt{\langle v, v\rangle}$ the length of a vector (Definition 7.6). ◁

Example 7.3 (A computation in $\mathbb{C}^2$). With $u = (1, i)$ and $v = (2,\ 1 - i)$ in $\mathbb{C}^2$,

$$\langle u, v\rangle = \overline{1}\cdot 2 + \overline{i}\cdot(1 - i) = 2 + (-i)(1 - i) = 2 + (-i - 1) = 1 - i,$$
$$\langle v, u\rangle = \overline{2}\cdot 1 + \overline{(1 - i)}\cdot i = 2 + (1 + i)i = 2 + (i - 1) = 1 + i.$$

So $\langle v, u\rangle = \overline{\langle u, v\rangle}$, as conjugate symmetry requires. ◁

Proposition 7.4 (Basic properties). In any inner product space, $\langle v, 0\rangle = \langle 0, v\rangle = 0$. Moreover, if $\langle u, x\rangle = \langle w, x\rangle$ for all x, then $u = w$.

Proof. By linearity in the second slot, $\langle v, 0\rangle = \langle v, 0\cdot 0\rangle = 0\cdot\langle v, 0\rangle = 0$, and by conjugate symmetry $\langle 0, v\rangle = \overline{\langle v, 0\rangle} = 0$. For the second claim: by conjugate-linearity in the first slot, $\langle u - w, x\rangle = 0$ for all x. Taking $x = u - w$ gives $\langle u - w, u - w\rangle = 0$. By positive definiteness $u - w = 0$, that is, $u = w$. ∎

Example 7.5 (Checking the axioms). We check the axioms for the standard product on $\mathbb{C}^n$.

(1) Conjugate symmetry: $\langle u, v\rangle = \sum_k \overline{u_k} v_k = \overline{\sum_k \overline{v_k} u_k} = \overline{\langle v, u\rangle}$.

(2) Linearity in the second slot follows from the distributive law for numbers.

(3) Positive definiteness: $\langle v, v\rangle = \sum_k |v_k|^2 \geq 0$, and this sum is zero only when every coordinate $v_k = 0$.

The inner product of functions from Example 7.2 is checked in the same way, with the sums replaced by integrals. Only positive definiteness needs one more step: if a continuous function f is nonzero at some point, then $|f|^2$ is positive on a small interval around that point, so $\int_a^b |f|^2\, dt > 0$. ◁

2 Norm, orthogonality, and the Pythagorean theorem

Definition 7.6 (Norm and orthogonality).

1. The *norm* of a vector v is $\|v\| = \sqrt{\langle v, v\rangle}$.
2. A vector with $\|v\| = 1$ is a *unit vector* (in physics, a *normalized state*).
3. Two vectors are *orthogonal*, $u \perp v$, if $\langle u, v\rangle = 0$.
4. Vectors $e_1, \dots, e_m$ are *orthonormal* if they are unit vectors and pairwise orthogonal, that is, $\langle e_i, e_j\rangle = \delta_{ij}$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$ (the *Kronecker delta*).

Orthonormal *bases* are studied in detail in Lecture 8.

The norm is *homogeneous*: for every scalar a,

$$\|av\| = \sqrt{\langle av, av\rangle} = \sqrt{\overline{a}\, a\, \langle v, v\rangle} = |a|\, \|v\|.$$

Moreover, by positive definiteness the norm is zero only when $v = 0$. The number $\|u - v\|$ is called the *distance* between the vectors u and v. Distances and convergence are studied in Lectures 13–14. By homogeneity, dividing a nonzero vector by its norm gives a unit vector: $\|v/\|v\|\| = \|v\|/\|v\| = 1$.

Example 7.7 (Normalizing a vector). The squared norm of the vector $v = (1, i, 1) \in \mathbb{C}^3$ is

$$\|v\|^2 = \langle v, v\rangle = |1|^2 + |i|^2 + |1|^2 = 3.$$

Dividing the vector v by its norm $\sqrt{3}$ gives the unit vector $\frac{1}{\sqrt{3}}(1, i, 1)$, which points in the same direction as v. ◁

Example 7.8 (Norms and inner products of functions). Take the space $\mathscr{C}[0, 1]$ with the inner product $\langle f, g\rangle = \int_0^1 \overline{f}\, g\, dt$. The squared norm of the function $f(t) = t$ is

$$\|f\|^2 = \int_0^1 t^2\, dt = \tfrac{1}{3},$$

so $\|f\| = 1/\sqrt{3}$. The functions 1 and t are not orthogonal, because $\langle 1, t\rangle = \int_0^1 t\, dt = \frac{1}{2} \neq 0$. In the next lecture the Gram–Schmidt process will replace them with the orthogonal pair 1 and $t - \frac{1}{2}$. On $[-1, 1]$ the same process gives the Legendre polynomials. ◁

The norm and orthogonality are linked by the Pythagorean theorem.

Theorem 7.9 (Pythagorean theorem).

1. If $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.
2. More generally, if the vectors $v_1, \dots, v_m$ are pairwise orthogonal, then $\|v_1 + \cdots + v_m\|^2 = \|v_1\|^2 + \cdots + \|v_m\|^2$.

Proof. Expanding the inner product in both slots gives

$$\|u + v\|^2 = \langle u + v, u + v\rangle = \|u\|^2 + \langle u, v\rangle + \langle v, u\rangle + \|v\|^2.$$

When $u \perp v$, both cross terms $\langle u, v\rangle = \langle v, u\rangle = 0$ vanish, and $\|u\|^2 + \|v\|^2$ remains. In the same way $\langle \sum_j v_j, \sum_k v_k\rangle = \sum_{j,k} \langle v_j, v_k\rangle$, and the terms with $j \neq k$ vanish by orthogonality. ∎

The inner product also lets us test linear independence.

Example 7.10 (Linear independence and the Gram matrix). The *Gram matrix* of vectors $v_1, \dots, v_m$ has entries $G_{jk} = \langle v_j, v_k \rangle$. The vectors are linearly independent if and only if the matrix G is invertible. Indeed, for coefficients $c = (c_1, \dots, c_m)$ put $w = \sum_k c_k v_k$. Pairing w with each v_j gives $\langle v_j, w \rangle = \sum_k G_{jk} c_k = (Gc)_j$, so $w = 0$ implies $Gc = 0$. Conversely, if $Gc = 0$, then $\|w\|^2 = \sum_j \overline{c_j} \langle v_j, w \rangle = c^\dagger G c = 0$, so $w = 0$ by positive definiteness. Hence the nonzero solutions of $Gc = 0$ are exactly the nontrivial linear relations among the v_k. For example, the Gram matrix of the vectors $v_1 = (1, 0)$ and $v_2 = (1, 1)$ in $\mathbb{R}^2$ is $G = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, and $\det G = 1 \neq 0$. So these vectors are linearly independent. ◁

Definition 7.11 (Orthogonal complement). The *orthogonal complement* of a subspace $U \subseteq V$ is the set

$$U^\perp = \{ v \in V : \langle u, v \rangle = 0 \text{ for all } u \in U \}.$$

It is also a subspace of V: it contains 0 (Proposition 7.4), and if $\langle u, v \rangle = \langle u, w \rangle = 0$, then $\langle u, av + bw \rangle = a\langle u, v \rangle + b\langle u, w \rangle = 0$ by linearity in the second slot.

We define the orthogonal complement here because only the inner product is needed to define it. We will use it in Lecture 8. There we will split V into the direct sum $U \oplus U^\perp$ and find a formula for the best approximation of a vector in the subspace U.

3 The Cauchy–Schwarz inequality

Every vector u can be split into two parts: one parallel to a given vector v, the other orthogonal to it (Figure 1). This splitting, together with the Pythagorean theorem, proves the main inequality of this lecture.

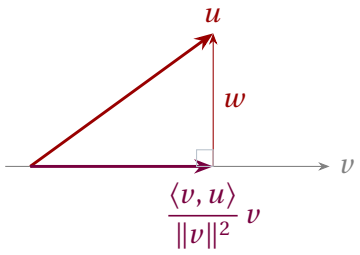


Figure 1: The orthogonal decomposition $u = \frac{\langle v,u \rangle}{\|v\|^2} v + w$ of the vector u, with $w \perp v$. We will apply the Pythagorean theorem to this right triangle.

Theorem 7.12 (Cauchy–Schwarz inequality). In an inner product space, for all u, v,

$$|\langle u, v \rangle| \leq \|u\| \, \|v\|.$$

Equality holds if and only if u, v are linearly dependent.

Proof. If $v = 0$, both sides of the inequality are zero, and the vectors u and 0 are linearly dependent. Assume from now on that $v \neq 0$, and write

$$u = \frac{\langle v, u \rangle}{\|v\|^2} v + w, \qquad w = u - \frac{\langle v, u \rangle}{\|v\|^2} v.$$

A direct check shows that $w \perp v$ (Figure 1):

$$\langle v, w \rangle = \langle v, u \rangle - \frac{\langle v, u \rangle}{\|v\|^2} \langle v, v \rangle = 0.$$

So the two summands are orthogonal, and by the Pythagorean theorem

$$\|u\|^2 = \frac{|\langle v,u\rangle|^2}{\|v\|^2} + \|w\|^2 \geq \frac{|\langle v,u\rangle|^2}{\|v\|^2}.$$

Multiplying by $\|v\|^2$ and taking square roots gives $|\langle v,u\rangle| \leq \|u\|\|v\|$. By conjugate symmetry $|\langle u,v\rangle| = |\langle v,u\rangle|$, which proves the inequality. Equality holds if and only if $\|w\| = 0$, that is, $w = 0$. This means that u is a multiple of v, that is, the vectors u and v are linearly dependent. ∎

Example 7.13 (Cauchy–Schwarz, checked). Take the vectors $u = (1, i)$ and $v = (2, 1-i)$ from Example 7.3. Their norms and the modulus of their inner product are

$$\|u\| = \sqrt{1+1} = \sqrt{2}, \quad \|v\| = \sqrt{4+2} = \sqrt{6} \quad \text{and} \quad |\langle u,v\rangle| = |1-i| = \sqrt{2}.$$

Then $\sqrt{2} \leq \sqrt{2}\cdot\sqrt{6} = 2\sqrt{3}$. The inequality is strict, because the vectors u and v are linearly independent. ◁

In a real space Cauchy–Schwarz lets us define the *angle* θ between nonzero vectors:

$$\cos\theta = \frac{\langle u,v\rangle}{\|u\|\,\|v\|}.$$

By Cauchy–Schwarz the right-hand side lies in $[-1,1]$, so such an angle exists.

Example 7.14 (Orthogonalizing a pair). The splitting in the proof of Cauchy–Schwarz gives a simple way to obtain an orthogonal pair. If the vectors v_1 and v_2 are linearly independent, then the vector

$$w_2 = v_2 - \frac{\langle v_1, v_2\rangle}{\|v_1\|^2} v_1$$

is orthogonal to v_1, just as $w \perp v$ in Figure 1. For example, in $\mathbb{R}^2$ take $v_1 = (1,0)$ and $v_2 = (1,1)$. Then

$$w_2 = (1,1) - 1\cdot(1,0) = (0,1).$$

Repeating this step for every basis vector gives the Gram–Schmidt process, which we study in Lecture 8. ◁

Example 7.15 (The closest point on a line). Let e be a unit vector. Which multiple ce is closest to the vector u? As in the proof of Cauchy–Schwarz, the error $u - \langle e,u\rangle\, e$ is orthogonal to e, so by the Pythagorean theorem $\|u - ce\|^2 = \|u - \langle e,u\rangle\, e\|^2 + |c - \langle e,u\rangle|^2$. This expression is smallest when $c = \langle e,u\rangle$. So the closest vector is $\langle e,u\rangle\, e$, the *orthogonal projection* of u onto the line in the direction of e. The least-squares method and the orthonormal expansions of Lecture 8 rest on this idea. ◁

4 Triangle inequality, parallelogram law, and polarization

Theorem 7.16 (Triangle inequality). In an inner product space, for all u, v,

$$\|u+v\| \leq \|u\| + \|v\|.$$

Proof. As in the proof of the Pythagorean theorem, expand $\|u+v\|^2$. The cross terms add up to $\langle u,v\rangle + \langle v,u\rangle = 2\,\mathrm{Re}\langle u,v\rangle$, because $\langle v,u\rangle = \overline{\langle u,v\rangle}$. Next we use $\mathrm{Re}\, z \leq |z|$ and Cauchy–Schwarz:

$$\|u+v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u,v\rangle + \|v\|^2 \leq \|u\|^2 + 2|\langle u,v\rangle| + \|v\|^2 \leq \big(\|u\| + \|v\|\big)^2.$$

Taking square roots gives the triangle inequality. ∎

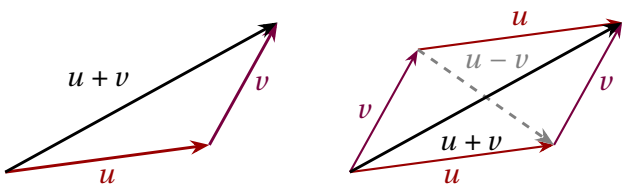


Figure 2: Left: the triangle inequality. The direct path $u+v$ is no longer than the two sides u and v: $\|u+v\| \le \|u\| + \|v\|$. Right: the parallelogram law. The squares of the two diagonals $u+v$ and $u-v$ add up to the squares of the four sides: $\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$.

Expanding $\|u \pm v\|^2$ gives two more useful identities: the parallelogram law, which relates the two diagonals in Figure 2 (right), and the polarization identity.

Proposition 7.17 (Parallelogram law and polarization identity). In an inner product space, for all u, v, the *parallelogram law*

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$$

holds. Moreover, in a complex inner product space the inner product can be recovered from the norm by the *polarization identity*

$$\langle u, v\rangle = \tfrac{1}{4} \sum_{\alpha \in \{1,-1,i,-i\}} \alpha \, \|v + \alpha u\|^2.$$

Proof. (1) *Parallelogram law.* Expanding $\|u \pm v\|^2 = \langle u \pm v, u \pm v\rangle$ gives

$$\|u \pm v\|^2 = \|u\|^2 + \|v\|^2 \pm 2\Re\langle u, v\rangle.$$

Adding the two equations (with the plus and with the minus sign) cancels the cross terms and gives the parallelogram law. (2) *Polarization identity.* Each $\alpha \in \{1, -1, i, -i\}$ has $|\alpha| = 1$, so the same expansion gives

$$\|v + \alpha u\|^2 = \|v\|^2 + \|u\|^2 + \alpha\langle v, u\rangle + \overline{\alpha}\,\langle u, v\rangle.$$

Multiplying by α and summing over all four values of α, we use $\sum_\alpha \alpha = 0$, $\sum_\alpha \alpha^2 = 0$ and $\sum_\alpha \alpha\overline{\alpha} = 4$. What remains is $\sum_\alpha \alpha \, \|v + \alpha u\|^2 = 4\langle u, v\rangle$. ∎

The polarization identity shows that the inner product is completely determined by the lengths of vectors. In a real space it simplifies to $\langle u, v\rangle = \frac{1}{4}\big(\|u+v\|^2 - \|u-v\|^2\big)$.

Example 7.18 (An angle, the parallelogram law, and polarization). In $\mathbb{R}^2$ take the vectors $u = (1, 0)$ and $v = (1, 1)$. They have $\langle u, v\rangle = 1$, $\|u\| = 1$, $\|v\| = \sqrt{2}$, so $\cos\theta = \frac{1}{\sqrt{2}}$ and $\theta = 45°$. Let us also check the parallelogram law:

$$\|u+v\|^2 + \|u-v\|^2 = \|(2,1)\|^2 + \|(0,-1)\|^2 = 5 + 1 = 6 = 2\|u\|^2 + 2\|v\|^2.$$

The real polarization identity recovers the inner product from these lengths: $\langle u, v\rangle = \frac{1}{4}\big(\|u+v\|^2 - \|u-v\|^2\big) = \frac{1}{4}(5-1) = 1$, the value computed directly. ◁

Example 7.19 (A norm with no inner product). The parallelogram law helps to recognize whether a measure of length can come from some inner product. The *max norm* $\|(x, y)\|_\infty = \max(|x|, |y|)$ on the plane (the general notion of a norm is defined in Lecture 13) violates it: for $u = (1, 0)$ and $v = (0, 1)$

$$\|u+v\|_\infty^2 + \|u-v\|_\infty^2 = 2 \neq 4 = 2\|u\|_\infty^2 + 2\|v\|_\infty^2.$$

Hence, by Proposition 7.17, no inner product can induce the norm $\|\cdot\|_\infty$. ◁

5 Dirac notation, amplitudes, and what comes next

Let us now translate the ideas of this lecture into the language of quantum mechanics. For this we need one more notion.

Definition 7.20 (Linear functional). A *linear functional* on V is a linear map $\varphi\colon V \to \mathbb{F}$ to the field of scalars.

For example, the trace $A \mapsto \operatorname{tr} A$ on $M_{n\times n}(\mathbb{C})$ and the integral $f \mapsto \int_a^b f\,dt$ on $\mathscr{C}[a,b]$ are linear functionals. Fixing a vector ϕ, we also get a linear functional from the inner product: by linearity in the second slot, the map $v \mapsto \langle \phi, v\rangle$ is linear.

Physicists write vectors as $|\psi\rangle$ and call them *kets*. To each vector ϕ Dirac assigns a *bra* $\langle\phi|$, which is exactly this linear functional:

$$\langle\phi|\colon |\psi\rangle \longmapsto \langle\phi \mid \psi\rangle = \langle \phi, \psi\rangle.$$

In Lecture 8 the Riesz theorem will show that on a finite-dimensional inner product space every linear functional is a bra.

The inner product is conjugate-linear in the first slot, so the correspondence $|\phi\rangle \mapsto \langle\phi|$ is conjugate-linear too:

$$a|\phi\rangle + b|\chi\rangle \mapsto \overline{a}\,\langle\phi| + \overline{b}\,\langle\chi|.$$

So the bracket $\langle\phi \mid \psi\rangle$ is linear in the ket $|\psi\rangle$ and conjugate-linear in the vector ϕ that labels the bra, exactly as Definition 7.1 requires.

Remark 7.21 (The Born rule). In quantum mechanics the state of a system is described by a unit vector $|\psi\rangle$, that is, $\langle\psi \mid \psi\rangle = 1$. Let $|\phi\rangle$ be another such state. The number $\langle\phi \mid \psi\rangle$ is called a *probability amplitude*. By the *Born rule*, the square of its modulus, $|\langle\phi \mid \psi\rangle|^2$, is the probability that a measurement finds the system, prepared in the state $|\psi\rangle$, in the state $|\phi\rangle$. You will study this rule in detail in the Quantum Mechanics course.

Remark 7.22 (Amplitudes are bounded). When $|\phi\rangle$ and $|\psi\rangle$ are unit vectors, Cauchy–Schwarz (Theorem 7.12) gives

$$|\langle\phi \mid \psi\rangle| \le 1.$$

So the modulus of an amplitude never exceeds one. Hence the Born-rule number $|\langle\phi \mid \psi\rangle|^2$ lies between zero and one, as a probability must. In Lecture 9 the same inequality will give the Heisenberg uncertainty relation.

Example 7.23 (An amplitude). Let

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big) \quad \text{and} \quad |\phi\rangle = |0\rangle,$$

where the states $|0\rangle$ and $|1\rangle$ are orthonormal: $\langle 0 \mid 0\rangle = \langle 1 \mid 1\rangle = 1$ and $\langle 0 \mid 1\rangle = 0$. By the Pythagorean theorem $\langle\psi \mid \psi\rangle = \tfrac12 + \tfrac12 = 1$, so $|\psi\rangle$ is a unit vector too. Then

$$\langle\phi \mid \psi\rangle = \tfrac{1}{\sqrt{2}}\big(\langle 0 \mid 0\rangle + \langle 0 \mid 1\rangle\big) = \tfrac{1}{\sqrt{2}},$$

so the probability of finding the system in the state $|0\rangle$ is $|\langle\phi \mid \psi\rangle|^2 = \tfrac12$. ◁

By a postulate of quantum mechanics, a nonzero vector and any nonzero multiple of it describe the same physical state. So we can always divide a vector by its norm, as in Example 7.7, and describe states by unit vectors. Even then, the unit vectors $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$, with $\theta \in \mathbb{R}$, describe the same state: the common factor $e^{i\theta}$, called a *phase*, changes no probability, because $|\langle\phi \mid e^{i\theta}\psi\rangle| = |\langle\phi \mid \psi\rangle|$.

The product in the opposite order, $|\psi\rangle\langle\phi|$, is no longer a number but an *operator*. It sends a vector $|\chi\rangle$ to the vector

$$|\psi\rangle\langle\phi \mid \chi\rangle,$$

that is, to a multiple of $|\psi\rangle$. Such an operator is called an *outer product*. If ψ and ϕ are nonzero, its rank is one: all its values are multiples of $|\psi\rangle$, and it sends $|\phi\rangle$ to the nonzero vector $\|\phi\|^2|\psi\rangle$. For a unit vector $|e\rangle$, the outer product $|e\rangle\langle e|$ sends $|u\rangle$ to $|e\rangle\langle e \mid u\rangle$, the part of u parallel to e (Figure 1 with $\|e\| = 1$). So $|e\rangle\langle e|$ is the orthogonal projection onto the line in the direction of e. In Lecture 8 we will build projections and resolutions of the identity from outer products.

Example 7.24 (An outer product as a matrix). In $\mathbb{C}^2$ take $|0\rangle = (1,0)$ and $|1\rangle = (0,1)$. We write kets as columns and bras as rows (conjugate transposes). Then the outer product

$$|0\rangle\langle 0| = \begin{pmatrix}1\\0\end{pmatrix}\begin{pmatrix}1 & 0\end{pmatrix} = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix}$$

is the projection onto the direction of $|0\rangle$:

$$|0\rangle\langle 0|\big(a\,|0\rangle + b\,|1\rangle\big) = a\,|0\rangle\langle 0 \mid 0\rangle + b\,|0\rangle\langle 0 \mid 1\rangle = a\,|0\rangle.$$

Adding the two outer products $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ gives the identity matrix:

$$|0\rangle\langle 0| + |1\rangle\langle 1| = \begin{pmatrix}1 & 0\\0 & 0\end{pmatrix} + \begin{pmatrix}0 & 0\\0 & 1\end{pmatrix} = I.$$

Such an expression for the identity matrix is called a *resolution of the identity*. ◁

Remark 7.25 (Expansions and expectation values). Let the vectors $|n\rangle$ form an orthonormal basis. Applying to $|\psi\rangle$ the *completeness relation* $\sum_n |n\rangle\langle n| = I$, which we prove in Lecture 8, gives the expansion

$$|\psi\rangle = \sum_n |n\rangle\langle n \mid \psi\rangle,$$

whose coefficients are the amplitudes. Its terms are pairwise orthogonal, so for a unit vector $|\psi\rangle$ the Pythagorean theorem gives $\sum_n |\langle n \mid \psi\rangle|^2 = 1$: the probabilities of all measurement outcomes add up to one. A measurable quantity is described by an operator A, called an *observable*. Its *expectation value* in the state $|\psi\rangle$ is $\langle A\rangle = \langle\psi|\,A\,|\psi\rangle$, the average of the measured values when the measurement is repeated many times in the same state. In Lecture 9 we will see that observables are the self-adjoint operators, and that their expectation values are real.

Example 7.26 (An expectation value). Let $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, and let the observable be the third Pauli matrix $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$. It describes the spin component along the z axis, and the possible measured values are its eigenvalues $+1$ and -1. Then $\sigma_z|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, so the expectation value is

$$\langle\sigma_z\rangle = \langle\psi|\,\sigma_z\,|\psi\rangle = \tfrac{1}{2}\big(\langle 0 \mid 0\rangle - \langle 0 \mid 1\rangle + \langle 1 \mid 0\rangle - \langle 1 \mid 1\rangle\big) = \tfrac{1}{2}(1-1) = 0.$$

The result can also be seen without computation. In Example 7.23 we found $\langle 0 \mid \psi\rangle = \frac{1}{\sqrt{2}}$, and in the same way $\langle 1 \mid \psi\rangle = \frac{1}{\sqrt{2}}$. So a measurement finds the system in the state $|0\rangle$ (value $+1$) and in the state $|1\rangle$ (value -1) with probability $\frac{1}{2}$ each, and the average of the values is zero. ◁

A measurement never confuses orthogonal states ($\langle\phi \mid \psi\rangle = 0$): a system in the state $|\psi\rangle$ is never found in the state $|\phi\rangle$. In Lecture 9 we prove that eigenvectors of an operator describing a measurable quantity, with different eigenvalues, are orthogonal. So different measurement outcomes correspond to orthogonal states. Orthogonal states can therefore be told apart by a suitable measurement, but not by every measurement, as the next example shows.

Example 7.27 (Qubit states). A quantum system whose states are the unit vectors of $\mathbb{C}^2$ is called a *qubit*. The vectors $|0\rangle$ and $|1\rangle$ are those of Example 7.24. Let

$$|+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad \text{and} \quad |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

These vectors are orthonormal. For example, $\langle + \mid - \rangle = \frac{1}{2}(\langle 0 \mid 0\rangle - \langle 1 \mid 1\rangle) = 0$, so a system in the state $|+\rangle$ is never found in the state $|-\rangle$. But the inner product of each of them with $|0\rangle$ equals $\frac{1}{\sqrt{2}}$, as in Example 7.23. So a measurement in the basis $|0\rangle, |1\rangle$ gives each outcome with probability $\frac{1}{2}$ in both cases. Such a measurement therefore does not tell $|+\rangle$ and $|-\rangle$ apart, although they differ only in the sign in front of $|1\rangle$. A measurement in the basis $|+\rangle, |-\rangle$ does tell them apart. ◁

In the next lecture we show that every finite-dimensional inner product space has an *orthonormal basis*, that is, a basis of pairwise orthogonal unit vectors. In such a basis the inner product becomes the familiar sum $\sum_k \overline{u_k} v_k$.

Summary of main concepts

An inner product is conjugate symmetric, positive definite, and linear in the second slot. Hence it is conjugate-linear in the first slot, that is, sesquilinear (the physics convention). The inner product defines the norm $\|v\| = \sqrt{\langle v, v\rangle}$ and orthogonality $\langle u, v\rangle = 0$, and the Pythagorean theorem links them. The Cauchy–Schwarz inequality $|\langle u, v\rangle| \le \|u\|\|v\|$ follows from the orthogonal decomposition and the Pythagorean theorem. The triangle inequality follows from it. The parallelogram law and the polarization identity show that the inner product is determined by the lengths of vectors. In Dirac notation $\langle\phi \mid \psi\rangle$ is linear in the ket $|\psi\rangle$ and conjugate-linear in ϕ. In quantum mechanics states are unit vectors, $|\langle\phi \mid \psi\rangle|^2$ is a probability, and a suitable measurement tells orthogonal states apart. These ideas are needed for the spectral theorem in Lecture 10.

- **Inner product, physics convention** — conjugate symmetric, linear in the second slot, positive definite; $\langle\phi \mid \psi\rangle$ is linear in the ket.
- **Norm and orthogonality** — $\|v\| = \sqrt{\langle v, v\rangle}$; $\langle u, v\rangle = 0$; the Pythagorean theorem.
- **Cauchy–Schwarz inequality** — $|\langle u, v\rangle| \le \|u\|\,\|v\|$; the modulus of an amplitude is at most one.
- **Triangle inequality, parallelogram law, and polarization** — properties of lengths and distances; $\langle\cdot,\cdot\rangle$ is recovered from $\|\cdot\|$.
- **Physical picture** — states are unit vectors, $|\langle\phi \mid \psi\rangle|^2$ is a probability, orthogonality means distinguishability.